

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)**REVIEW TEST****27-12-2020****JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]****PAPER-2****Time : 3 Hours****Maximum Marks : 186****READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

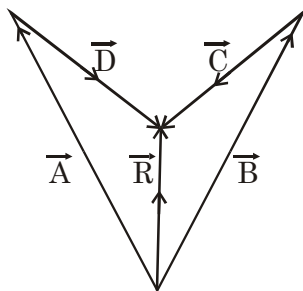
SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. A dimensionally correct equation is given by $E = \sqrt{\frac{P^4}{2m^2}} + k\sqrt{a}$ in which E is energy, P is momentum, m is mass and k is dimensionless then :-
- (A) Dimensions of a^2 is same as energy
- (B) Dimensions of $\sqrt{a}$ is same as energy
- (C) Dimensions of $(m^2a)^{1/4}$ is same as momentum
- (D) $\frac{E^2}{a}$ is dimensionless

Space for Rough Work

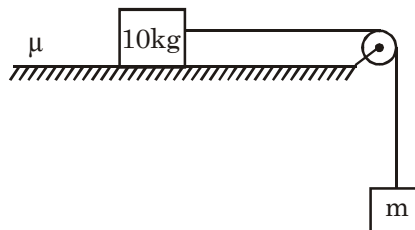
2. Which of the following option(s) is/are correct for the vectors given in figure ?



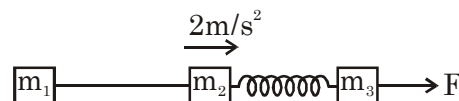
- (A) $\vec{A} = \vec{R} - \vec{D}$ (B) $\vec{A} - \vec{B} = \vec{C} - \vec{D}$
 (C) $\vec{B} + \vec{C} = \vec{R}$ (D) None of these
3. A ball is dropped from point A at a height of 20m from ground, it moves under gravity and collides at point B which is vertically below from A. After collision ball reaches to point C which is 5m above the ground and ball was in contact with ground for 0.5 sec.
 (take $g = 10 \text{ m/s}^2$)
 (A) Time to reach from A to B is 2 sec.
 (B) Time to reach from B to C is 1 sec.
 (C) Velocity just before it hits at B is 20 m/s
 (D) Velocity just after it leaves from point B is 10m/s
4. Two cars A & B start from same point at $t = 0$. A is moving with a uniform velocity of 3m/s towards east and B is moving with a uniform velocity of 4 m/s towards north direction.
 (A) Relative speed of A w.r.t. B is 5 m/s.
 (B) Distance between A & B after 3 sec is 15 m.
 (C) Relative velocity of A w.r.t. B is 5 m/s in 53° south of east.
 (D) Relative velocity of B w.r.t. A is 5 m/s in 37° west of north.

Space for Rough Work

5. If coefficient of friction on the surface is μ , pulley and string between blocks of mass 10kg and m are ideal, then choose the correct option(s).



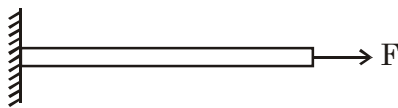
- (A) If $\mu = 0.5$ and $m = 10$ kg, blocks move.
 (B) If $\mu = 0.5$ and $m = 4$ kg, blocks move.
 (C) If $\mu = 1.5$ and $m = 14$ kg, blocks do not move
 (D) If $\mu = 1.5$ and $m = 20$ kg, blocks do not move.
6. In the given figure $m_1 = 1$ kg, $m_2 = 2$ kg, $m_3 = 3$ kg, $F = 18$ N and spring constant is 100 N/m then choose the correct options. (Assume string and spring are ideal)



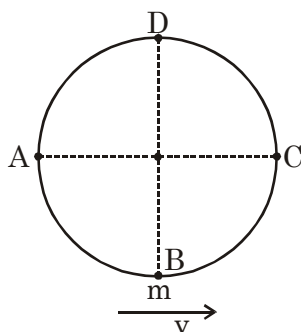
- (A) Acceleration of m_1 is 2 m/s^2
 (B) Tension in string is 2N
 (C) Force applied by spring on block of mass m_2 is 6N
 (D) Acceleration of block of mass m_3 is 4 m/s^2

Space for Rough Work

7. A rod of length L and cross-section area A is fixed with rigid support in gravity free space and a force F is applied at the other end. (Young's modulus of rod is Y)



- (A) Tension in rod is F .
- (B) Elongation in rod is $\frac{FL}{AY}$
- (C) Elastic potential energy stored in rod is $\frac{F^2L}{2AY}$
- (D) Tension in rod is $\frac{F}{2}$
8. A particle of mass m attached with an ideal string of length ℓ , can perform verticle circular motion. A and C are the points where string becomes horizontal. If velocity of mass at lowest point is V then :-



- (A) If $V = \sqrt{g\ell}$; it can not reach to the point C.
- (B) If $V = \sqrt{6g\ell}$; it performs verticle complete circular motion.
- (C) If $V = \sqrt{3g\ell}$; string slacks between C and D.
- (D) If $V = \sqrt{2g\ell}$; tension in string at lowest point is $3mg$.

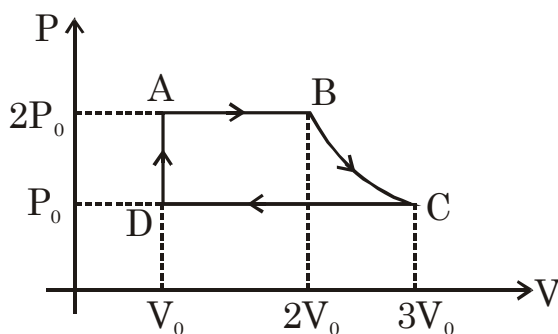
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

An ideal monoatomic gas follows a cyclic process. The pressure of gas (P) changes with volume of gas (V) as shown in diagram. All process are quassi-static. Process B → C is isothermal.



List-I

- (I) Change in internal energy of process A → B is
 (II) Work done by gas in process D → A is
 (III) Change in internal energy of process B → C is
 (IV) Heat given to the gas in process C → D is

List-II

- (P) Positive
 (Q) Negative
 (R) Zero
 (S) $3P_0V_0$
 (T) $-\frac{20}{3}P_0V_0$

- (U) $\frac{20}{3}P_0V_0$

(A) I → P,S; II → R, III → R, IV → Q,T

(B) I → P,U, II → T; III → Q, IV → R,S

(C) I → Q,T; II → P,U; III → R, IV → R

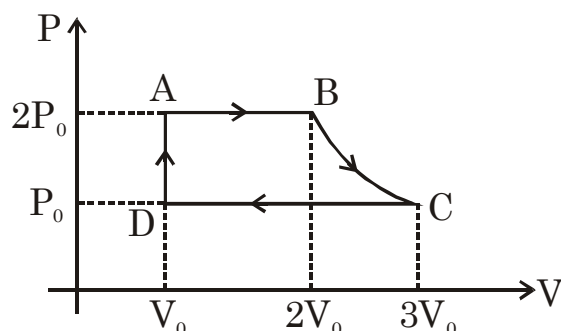
(D) I → P,S; II → R, III → Q,T; IV → R

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

An ideal monoatomic gas follows a cyclic process. The pressure of gas (P), changes with volume of gas (V) as shown in diagram. All process are quasi-static. Process B → C is isothermal.

$$[\ln(1.5) = 0.4; \sqrt{2} = 1.4; \sqrt{3} = 1.73]$$



- | List-I | List-II |
|---|------------------------------------|
| (I) Efficiency of cyclic process is | (P) 0.70 |
| (II) Ratio of rms speed of gas at state A and rms speed at state B is | (Q) 0.13 |
| (III) Ratio of rms speed of gas at state B and state C is | (R) 1 |
| (IV) Ratio of most probable speed of gas at state A and rms speed at state C is | (S) 0.57 |
| | (T) 1.51 |
| | (U) 0.95 |
| (A) I → S, II → U, III → T, IV → P | (B) I → R, II → Q, III → P, IV → S |
| (C) I → P, II → S, III → Q, IV → T | (D) I → Q, II → P, III → R, IV → S |

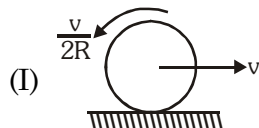
Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

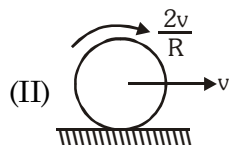
Four different situations of a moving disc are shown in List-I and predictions about its final motion and forces acting on it are given in list-II.

List-I

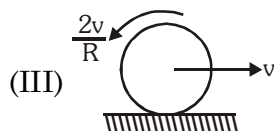
List-II



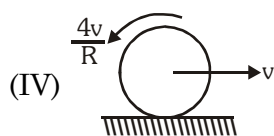
(P) Finally disc will roll along the initial direction of velocity (v)



(Q) Finally, disc will roll in direction opposite to the initial direction of velocity (v)



(R) Finally, disc stops



(S) Initially friction force acts in the direction opposite to that of initial velocity

(T) Friction in direction same as that of initial velocity.

(U) No friction act on the disc.

- (A) I → P,Q; II → R,S; III → P,S; IV → Q,R
 (B) I → P,U; II → P,S; III → Q,T; IV → Q,T
 (C) I → P,S; II → P,T; III → R,S; IV → Q,S
 (D) I → Q,R; II → P,T; III → S,U; IV → P,Q

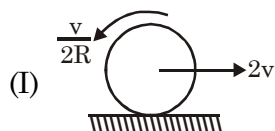
Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

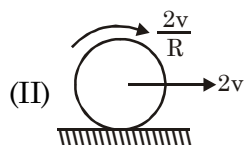
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List-I

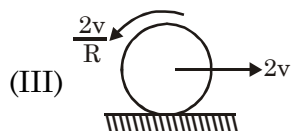
List-II



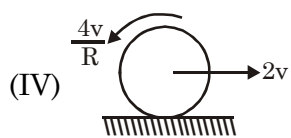
(P) Finally disc will roll along the initial direction of velocity (v)



(Q) Finally, disc will roll in direction opposite to the initial direction of velocity (v)



(R) Finally, disc stops



(S) Initially friction force acts in the direction opposite to that of initial velocity

(T) Friction in direction same as that of initial velocity.

(U) No friction act on the disc.

- (A) I $\rightarrow$ P,S; II $\rightarrow$ P,U; III $\rightarrow$ P,S; IV $\rightarrow$ R, S
 (B) I $\rightarrow$ P,U; II $\rightarrow$ P,S; III $\rightarrow$ Q,T; IV $\rightarrow$ Q,T
 (C) I $\rightarrow$ P,S; II $\rightarrow$ P,T; III $\rightarrow$ R,S; IV $\rightarrow$ Q,S
 (D) I $\rightarrow$ Q,R; II $\rightarrow$ P,T; III $\rightarrow$ S,U; IV $\rightarrow$ P,Q

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

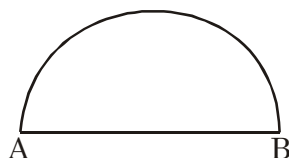
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS. **For Example** : If answer is -77.25, 5.2 then fill the bubbles as follows.

										+										
●	●	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	

										-										
●	●	●	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

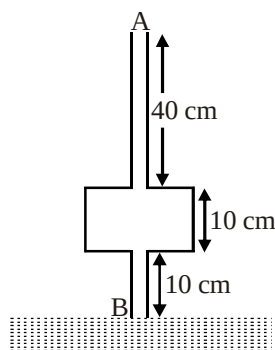
- Two wires of same material and same cross section area are connected between A & B, with one being semicircular and other one straight. Temperature of A is 20°C and temperature of B is 100°C. If length of straight wire is 14m, cross section area of wires is 2 cm² and thermal conductivity is 18000 W/mk then total rate of heat transfer in SI unit is.



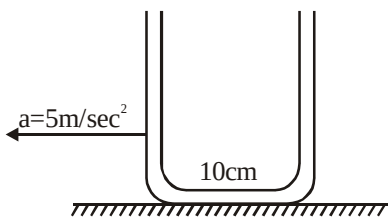
- A spherical ball of radius 7m is made up of a material having linear expansion coefficient $\alpha = 2 \times 10^{-5}$ m/s and bulk modulus 10^4 Pa. If ball is heated by 20°C then calculate the needed pressure (Pa) to keep its volume constant.

Space for Rough Work

3. In a flow calorimeter, water gets in at 20°C with mass flow rate of 4 g/s . Calculate the rate of heat (w) given to the calorimeter if water comes out of the calorimeter at 80°C . (take $1\text{J} = 4.2\text{ cal}$)
4. A ball is dropped from a height of 125m . After first collision with ground ball reaches to maximum height 100 m . If maximum heights achieved by the ball after 2^{nd} and 4^{th} collision are $-h_2$ and h_4 respectively then calculate $\frac{h_2}{h_4}$.
5. Water (density 1gm/cc) is to be sucked upto point A. The area of the narrow tube is 1cm^2 and the wide portion is 100 cm^2 . In this situation, minimum work required to be done is w_1 . When the tube is inverted and water is to be sucked upto point B, the minimum work required to be done is w_2 . Find $(w_2 - w_1)$ in Joule. Fill $100(w_2 - w_1)$ in OMR sheet.



6. The horizontal part of the U-shaped tube shown in the figure is 10 cm , and its vertical branches are long enough. The cross section of the tube is 1 cm^2 . First, 20 cm^3 water is poured into the tube, then 20 cm^3 oil is poured into the left branch. (density of water = 1000 kg/m^3 , density of oil = 800 kg/m^3). The whole system is accelerated towards the left, at an acceleration of $a = 5\text{ m/sec}^2$. Determine the difference between the level of water and the level of oil (in cm).



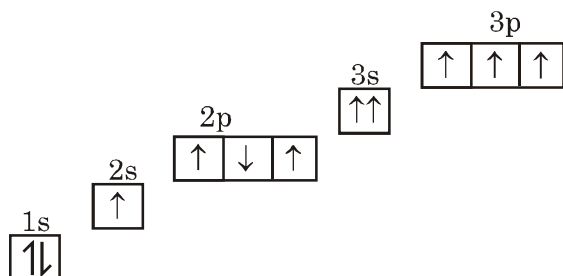
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Consider the following electronic configuration.



The rule(s) violated is/are

- (A) Aufbau's rule (B) Hund's rule (C) Pauli's rule (D) No rule is violated

Space for Rough Work

2. The silicate compound(s) in which all atoms of anionic part in same plane and share atleast two oxygen per tetrahedral unit is/are.
 (A) $(\text{Mg,Fe})_7\text{Si}_8\text{O}_{22}(\text{OH})_2$ (B) $(\text{Mn,Ca})_5\text{Si}_5\text{O}_{15}$
 (C) $\text{Ba}_2\text{Al}_2\text{Si}_2\text{O}_8$ (D) None
3. Dinitrogen tetroxide is often a component in smog, it undergoes three different types of self dissociation according to the following -

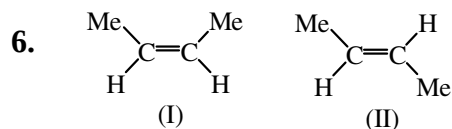
$$\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2 \quad ; K_{\text{eq}} = 10^{-2}$$

$$\text{N}_2\text{O}_4 \rightleftharpoons \text{NO}^+ + \text{NO}_3^- \quad ; K_{\text{eq}} = 10^{-7}$$

$$\text{N}_2\text{O}_4 \rightleftharpoons \text{NO}_2^+ + \text{NO}_2^- \quad ; K_{\text{eq}} = 10^{-22}$$
 At 303 K, assuming no other reaction, which are true regarding concentration of species at 303K at equilibrium (initially only N_2O_4 was there)
 (A) $[\text{N}_2\text{O}_4] > [\text{NO}_2]$ (B) $[\text{N}_2\text{O}_4] > [\text{NO}^+]$
 (C) $[\text{NO}^+] = [\text{NO}_3^-] = [\text{NO}_2^+] = [\text{NO}_2^-]$ (D) $[\text{NO}_3^-] > [\text{NO}_2^-]$
4. Some amount of a gas is taken in a closed rigid container at a certain pressure. Identify correct statement(s)
 (A) If pressure of the gas is increased twice, then mean free path of the gas is reduced to half of its initial value
 (B) If pressure of the gas is increased 16 times, then the collision frequency becomes 4 times of its initial value
 (C) If temperature of the gas is halved, then collision number is reduced to 0.707 of its initial value.
 (D) If temperature is halved, then mean free path is also halved for the above given gas

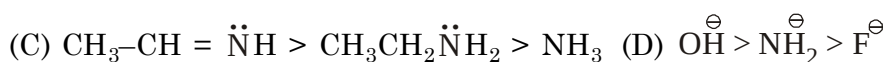
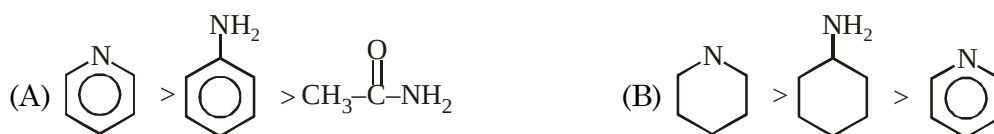
Space for Rough Work

5. A vessel contains O_2 gas saturated with water vapours at a total pressure of 2 atm at $37^\circ C$. The volume of the vessel is reduced to one-third of its initial volume at same temperature. (Given : aqueous tension at $37^\circ C = 76$ mm of Hg) Then finally ,
- (A) Total pressure = 6 atm (B) Total pressure = 5.8 atm
(C) Pressure of O_2 gas = 6 atm (D) Pressure of O_2 gas = 5.7 atm



Select the correct option related with above compound -

- (A) dipole moment of I > II (B) Stability of I < II
(C) Boiling point I > II (D) Melting point I < II
7. Correct order of basic strength in aq. solution is/are :



8. Which of the following are chiral object.

- (A) Screw (B) Fan
(C) Kids tricycle (D) Hockey stick

Space for Rough Work

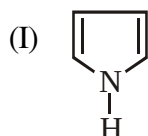
SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

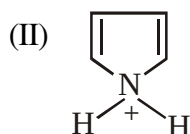
Answer the following by appropriately matching the lists based on the information given in the paragraph.

9. List-I

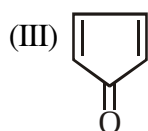
List-II



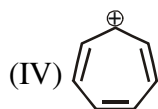
(P) Aromatic



(Q) Non aromatic



(R) Antiaromatic



(S) Homocyclic

(T) Heterocyclic

(U) Unstable at room temperature

Which of the following options has the **CORRECT** combination considering List-I and List-II?

(A) (IV), (R)

(B) (III), (U)

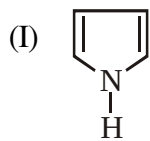
(C) (II), (S)

(D) (I), (Q)

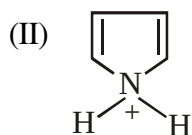
Space for Rough Work

10. List - I

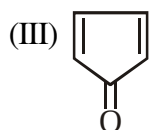
List- II



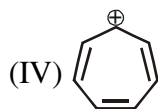
(P) Aromatic



(Q) Non aromatic



(R) Antiaromatic



(S) Homocyclic

(T) Heterocyclic

(U) Unstable at room temperature

Which of the following options has the **INCORRECT** combination considering List-I and List-II?

(A) (IV), (R)

(B) (III), (U)

(C) (II), (T)

(D) (I), (P)

Space for Rough Work

11. Consider a Bohr's Model for a single e^- species, Frequency of revolution of electron in n^{th} orbit = f , Time period of revolution of electron in n^{th} orbit = T , Energy of electron in n^{th} orbit = E_n , Atomic number = Z , Orbit number = n , $V_n = \text{P.E.}$; $K_n = \text{K.E.}$, $r_n = \text{radius of } n^{\text{th}} \text{ orbit}$

List-I

- (I) 0
(II) -2
(III) ∞
(IV) 1

List-II

- (P) V_n / K_n
(Q) $f \times T$
(R) E_n / V_n
(S) $\frac{E_2(\text{He}^+)}{E_3(\text{Li}^{2+})}$
(T) Last line of Paschen series corresponds to $n = ?$ to $n = 3$
(U) $\frac{r_1(\text{H})}{r_2(\text{Be}^{3+})}$

Which of the following options has the **CORRECT** combination considering List-I and List-II ?

- (A) (I), (S) (B) (II), (P) (C) (III), (Q) (D) (IV), (P)

Space for Rough Work

12. Consider a Bohr's Model for a single e^- species, Frequency of revolution of electron in n^{th} orbit = f , Time period of revolution of electron in n^{th} orbit = T , Energy of electron in n^{th} orbit = E_n , Atomic number = Z , Orbit number = n , V_n = P.E. ; K_n = K.E. , r_n = radius of n^{th} orbit

List-I

- (I) 0
(II) -2
(III) ∞
(IV) 1

List-II

- (P) V_n / K_n
(Q) $f \times T$
(R) E_n / V_n
(S) $\frac{E_2(\text{He}^+)}{E_3(\text{Li}^{2+})}$
(T) Last line of Paschen series corresponds to $n = ?$ to $n = 3$
(U) $\frac{r_1(\text{H})}{r_2(\text{Be}^{3+})}$

Which of the following options has the **INCORRECT** combination considering List-I and List-II ?

- (A) (IV), (Q) (B) (IV), (U) (C) (III), (T) (D) (II), (R)

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

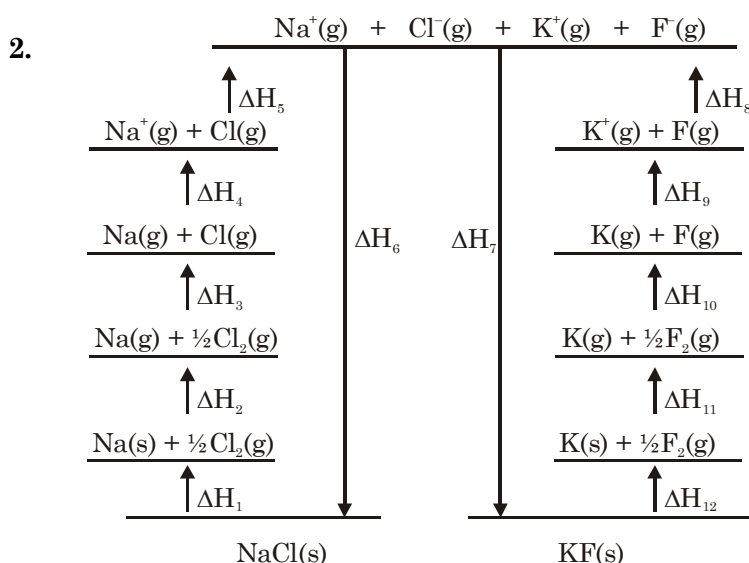
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+					-				
●	●	○	○	○	●	●	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	●
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	●	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

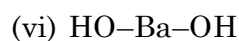
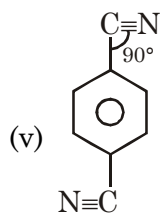
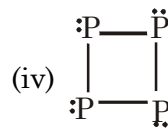
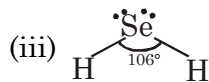
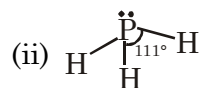
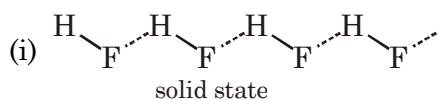
- The initial molar concentration of A and B are 0.80 mol/L each. The reaction starts to proceed as $A+B \rightleftharpoons C+D$ and attains equilibrium. At equilibrium, the molar concentration of C is 0.6 mole/L. The value of K_c is.



The number of enthalpies that are positive in the above diagram is.

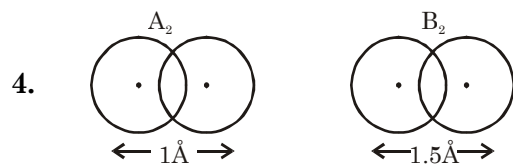
Space for Rough Work

3. Some structures are given below



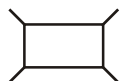
The number of **INCORRECT** structures is.

Space for Rough Work



The bond length of A–B bond is _____ (in Å) ($|\chi_A - \chi_B| = 0.36$)

5. Number of specie(s) among the following that are paramagnetic.
 Li^+ , Fe^{2+} , Cu^+ , V^{3+} , Sc^{3+} , O^- , Cl^+ , Zn^{2+}
6. Total numbers of geometrical isomers in following compounds.



Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Integral value(s) of 'a' such that the quadratic equation $x^2 + ax + a + 1 = 0$ has integral roots is/are
 (A) 1 (B) 5 (C) -1 (D) 3
2. Consider ΔABC of sides 3, 5 and 6. Altitudes AD, BE and CF drawn from vertices A, B and C on sides BC, CA and AB respectively are concurrent at H. If $\frac{(HB)(HC)(BC)}{\text{Area}(\Delta HBC)} = \frac{p}{\sqrt{q}}$ where p and q are coprime, then answer the following
 (A) $p = 35$ (B) $p = 45$ (C) $q = 14$ (D) $q = 64$
3. If the lines $ax + 4y + a^2 = 0$, $x - 2y - 6 = 0$ and $3x + y - 4 = 0$ are concurrent, then a can be
 (A) $a = 2$ (B) $a = -2$ (C) $a = 4$ (D) $a = -4$
4. Equation of common tangents to $y^2 = 4(x - 1)$ and $x^2 + y^2 = 1$ is-
 (A) $\sqrt{3}y = x + 2$ (B) $\sqrt{3}y + x + 2 = 0$ (C) $x = 1$ (D) $y = 1$

Space for Rough Work

5. Let a, b, c are different integers less than or equal to 10. The arithmetic mean of a and b is 9. The geometric mean of a and c is $6\sqrt{2}$. The harmonic mean of b and c is equal to
- (A) $\frac{18}{19}$ (B) $\frac{90}{19}$ (C) $\frac{100}{19}$ (D) $\frac{180}{19}$
6. Let $f(x) = \begin{vmatrix} x & 1 & -3 \\ 2 & 2 & 1 \\ \frac{1}{x-1} & 0 & \frac{1}{2} \end{vmatrix}$. The minimum value of $f(x)$ (given $x > 1$), is
- (A) $2\sqrt{2}$ (B) $2\sqrt{3}$ (C) 2 (D) 4
7. The parabola $y = 4 - x^2$ has vertex P. It intersects x-axis at A and B (where abscissa of B is positive). If the parabola is translated from its initial position to a new position by moving its vertex along the line $y = x + 4$, so that it intersects x-axis at B and C, then abscissa of C will be:
- (A) 2 (B) 4 (C) 6 (D) 8
8. All possible 120 permutations of WDSMC are arranged in dictionary order, as if each were an ordinary five-letter word. The last letter of the 86th word in the list, is
- (A) W (B) D (C) M (D) C

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I	List-II
(I) The value of $\left[\frac{2009!+2006!}{2008!+2007!} \right] - 2008$ ([.] denotes greatest integer function.)	(P) 0
(II) If the distance of a point P on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ from the centre is 2, P has eccentric angle $\theta \in (0, 2\pi)$ then $[\theta]$ is ([.] denotes greatest integer function)	(Q) 1
(III) If eccentricity of conjugate hyperbola of the given hyperbola is e' then value of $4e'$ is where eccentricity of hyperbola $e = 5/3$	(R) 2
(IV) Distance of point P(α, β) from (2,3), if the tangents to the curve $y^2 = 4x$ from point P are perpendicular, is	(S) 3 (T) 4 (U) 5
Which of the following is the only CORRECT combination ?	
(A) (I), (P)	(B) (II), (Q)
(C) (III), (R)	(D) (IV), (U)

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I		List-II
(I) The value of $\left[\frac{2009! + 2006!}{2008! + 2007!} \right] - 2008$	(P)	0
([.] denotes greatest integer function.)		
(II) If the distance of a point P on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ from the centre is 2, P has eccentric angle $\theta \in (0, 2\pi)$ then $[\theta]$ is ([.] denotes greatest integer function)	(Q)	1
(III) If eccentricity of conjugate hyperbola of the given hyperbola is e' then value of $4e'$ is where eccentricity of hyperbola $e = 5/3$	(R)	2
(IV) Distance of point $P(\alpha, \beta)$ from $(2, 3)$, if the tangents to the curve $y^2 = 4x$ from point P are perpendicular, is	(S)	3
	(T)	4
	(U)	5
Which of the following is the only CORRECT combination ?		
(A) (I), (U)	(B) (II), (T)	(C) (III), (U)
		(D) (IV), (U)

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I

List-II

- | | | | |
|-------|--|-----|---|
| (I) | For $0 < \theta < \frac{\pi}{2}$, if $\cos \theta = \frac{a}{b+c}$, where a, b, c are the sides of ΔABC then $\sin \theta = \frac{k\sqrt{bc}}{b+c} \cos \frac{A}{2}$. The value of k equals | (P) | 0 |
| (II) | The number of integral value(s) of 'a' for which the point P(a, a^2) lies completely inside the triangle formed by the lines $x = 0$, $y = 0$ and $x + 2y = 3$ equals | (Q) | 1 |
| | | (R) | 2 |
| | | (S) | 3 |
| (III) | If $3^{x-1} + 3^{x-2} + 3^{x-3} + 3^{x-4} + \dots \infty = 2 \left(5^2 + 5 + 1 + \frac{1}{5} + \dots \infty \right)$ then $x = 3 \log_3 N$. The value of N equals | (T) | 4 |
| (IV) | Number of straight lines passing through (3, 6) and forming a triangle of area 36 square unit with the co-ordinate axes, is | (U) | 5 |

Which of the following is the only **CORRECT** combination ?

- (A) (I), (P) (B) (II), (Q) (C) (III), (R) (D) (IV), (S)

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I

List-II

- | | | | |
|-------|--|-----|---|
| (I) | For $0 < \theta < \frac{\pi}{2}$, if $\cos \theta = \frac{a}{b+c}$, where a, b, c are the sides of ΔABC then $\sin \theta = \frac{k\sqrt{bc}}{b+c} \cos \frac{A}{2}$. The value of k equals | (P) | 0 |
| (II) | The number of integral value(s) of 'a' for which the point P(a, a ²) lies completely inside the triangle formed by the lines x = 0, y = 0 and x + 2y = 3 equals | (Q) | 1 |
| | | (R) | 2 |
| | | (S) | 3 |
| (III) | If $3^{x-1} + 3^{x-2} + 3^{x-3} + 3^{x-4} + \dots \infty = 2 \left(5^2 + 5 + 1 + \frac{1}{5} + \dots \infty \right)$ then $x = 3 \log_3 N$. The value of N equals | (T) | 4 |
| (IV) | Number of straight lines passing through (3, 6) and forming a triangle of area 36 square unit with the co-ordinate axes, is | (U) | 5 |

Which of the following is the only CORRECT combination ?

- (A) (I), (Q) (B) (II), (T) (C) (III), (U) (D) (IV), (P)

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+					-				
●	●	○	○	○	●	●	●	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	●
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	●	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let α_1, β_1 are the roots of $x^2 - 6x + p = 0$ and α_2, β_2 are the roots of $x^2 - 54x + q = 0$. If $\alpha_1, \beta_1, \alpha_2, \beta_2$ form an increasing G.P., then find the value of $(q - p)$.
- A circle $x^2 + y^2 + 4x - 2\sqrt{2}y + c = 0$ is the director circle of circle S_1 and S_1 is the director circle of circle S_2 and so on. If the sum of radii of all these circles is 2, then the value of c is equal to $\sqrt{n}$ where $n \in \mathbb{N}$. Find the value of n .
- A list is made of all the 3 digit numbers whose digits are 1, 2, 3, 4, 5 with no repetition. If the sum of all the numbers is 10 N, then find the value of N .

Space for Rough Work

4. If $\log_b 3 = 4$ and $\log_{b^2} 27 = \frac{3a}{2}$ then the value of $\frac{a^2 - b^4}{3}$ is
5. If the points with coordinates $(a, 0)$ and $(0, b)$ are equidistant from the point $(1, 4)$ and $(9, 0)$ then $\frac{(a-b)}{10}$ equals
6. Nine people sit around a round table. The number of ways of selecting four of them such that they are not from adjacent seats, is
-

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	32
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator